

**REPORT CLASSIFICATION**

|                     |                    |
|---------------------|--------------------|
| Classification Type | Full Form Analysis |
|---------------------|--------------------|

**DATE / TIME**

|                    |           |
|--------------------|-----------|
| Date of Occurrence | 2001-03   |
| Local Time Of Day  | 1201-1800 |

**PLACE**

|                                             |             |
|---------------------------------------------|-------------|
| Locale Reference                            | Airport     |
| Locale                                      | ZZZ.Airport |
| State                                       | US          |
| Relative Position - Distance Nautical Miles | 0           |
| Altitude - AGL                              | 0           |

**ENVIRONMENT**

|       |          |
|-------|----------|
| Light | Daylight |
|-------|----------|

**AIRCRAFT / EQUIPMENT X**

|                                                 |             |
|-------------------------------------------------|-------------|
| Aircraft Operator                               | Air Carrier |
| Make Model Name                                 | B737-300    |
| Crew Size                                       | 2           |
| Operating Under FAR Part                        | Part 121    |
| Mission                                         | Passenger   |
| Flight Phase                                    | Parked      |
| Maintenance Status - Maintenance Deferred       | Y           |
| Maintenance Status - Released For Service       | Y           |
| Maintenance Status - Maintenance Items Involved | Repair      |

**COMPONENT 1**

|                    |                     |
|--------------------|---------------------|
| Aircraft Reference | X                   |
| Aircraft Component | Nacelle/Pylon Skin  |
| Problem            | Improperly Operated |

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**COMPONENT 2**

|                    |                              |
|--------------------|------------------------------|
| Aircraft Reference | X                            |
| Aircraft Component | Minimum Equipment List (MEL) |
| Problem            | Improperly Operated          |

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**PERSON 1**

|                                     |               |
|-------------------------------------|---------------|
| Location Of Person                  | Company       |
| Reporter Organization               | Air Carrier   |
| Function - Maintenance              | Technician    |
| Qualification - Maintenance         | Airframe      |
| Qualification - Maintenance         | Powerplant    |
| Experience - Maintenance Technician | 23 yrs        |
| ASRS Report Number                  | 505831        |
| Human Factors                       | Time Pressure |

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**PERSON 2**

|                             |             |
|-----------------------------|-------------|
| Location Of Person          | Company     |
| Reporter Organization       | Air Carrier |
| Function - Maintenance      | Technician  |
| Qualification - Maintenance | Airframe    |

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**PERSON 3**

|                             |                           |
|-----------------------------|---------------------------|
| Aircraft Reference          | X                         |
| Location Of Person          | Company                   |
| Reporter Organization       | Air Carrier               |
| Function - Flight Crew      | Captain                   |
| Function - Flight Crew      | Pilot Flying              |
| Qualification - Flight Crew | Air Transport Pilot (ATP) |

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**PERSON 4**

|                             |                 |
|-----------------------------|-----------------|
| Location Of Person          | Company         |
| Reporter Organization       | Air Carrier     |
| Function - Maintenance      | Lead Technician |
| Qualification - Maintenance | Airframe        |
| Qualification - Maintenance | Powerplant      |

**EVENTS**

|                   |                                                                    |
|-------------------|--------------------------------------------------------------------|
| Anomaly           | Aircraft Equipment Problem - Critical                              |
| Anomaly           | Deviation / Discrepancy - Procedural - FAR                         |
| Anomaly           | Deviation / Discrepancy - Procedural - Maintenance                 |
| Anomaly           | Deviation / Discrepancy - Procedural - MEL / CDL                   |
| Anomaly           | Deviation / Discrepancy - Procedural - Published Material / Policy |
| Detector - Person | Flight Crew                                                        |
| Result - General  | Flight Cancelled / Delayed                                         |
| Result - General  | Maintenance Action                                                 |
| Result - General  | None Reported / Taken                                              |
| Result - Aircraft | Aircraft Damaged                                                   |

**ASSESSMENTS**

|                                   |                      |
|-----------------------------------|----------------------|
| Contributing Factors / Situations | Aircraft             |
| Contributing Factors / Situations | Chart Or Publication |
| Contributing Factors / Situations | Human Factors        |
| Contributing Factors / Situations | Manuals              |
| Primary Problem                   | Chart Or Publication |

**NARRATIVE 1**

ON WALKAROUND, THE FLC RPTED A SMALL DENT IN THE #2 ENG NOSE COWL. DIMENSIONS WERE 1.5 INCHES DIAMETER, 3/32 INCHES DEEP, BEYOND THE LIMITS OF THE STRUCTURAL REPAIR MANUAL (SRM). I DIRECTED THE MECH TO REWORK THE DENT TO MEET THE LIMITS OF THE SRM WHICH HE DID. THERE IS A DISCREPANCY IN THE SRM WHERE ATA CHAPTER 54 SAYS TO REWORK DENTS TO SRM LIMITS AND CHAPTER 51 ATA SAYS DENTS CAN BE REWORKED OF METAL .063 OR LESS. METAL ON THE NOSE COWL IS .100. THE SRM DOES NOT SAY YOU CANNOT REWORK DENTS OF THAT THICKNESS, BUT IT DOES NOT SAY YOU CAN. THE DENT HAS BEEN REWORKED AND IS WITHIN LIMITS. THE ENGINEERING DEPT IS GOING TO CLARIFY THE MANUALS AFTER CONFERRING WITH BOEING. SUPPLEMENTAL INFO FROM ACN 506195: IT WAS DETERMINED THAT THE DAMAGE WAS OUT OF LIMITS PER SRM 54-10-01 PAGE 104. THE ACTUAL DAMAGE WAS 7 INCHES LONG X 1 1/2 INCHES WIDE X 3/32 INCHES DEEP. TO CONFORM TO SRM LIMITS, THE DAMAGE NEEDED TO HAVE A FACTOR OF 30, MEANING THE DEPTH WAS 1/13 OF THE MINOR WIDTH. WE ATTEMPTED TO GET AN

ENGINEERING VARIANCE AUTH TO DEFER THE DAMAGE, BUT WE WERE UNSUCCESSFUL. COMMERCIAL MAINT ORGANIZATION SUGGESTED TO EITHER LESSEN THE DEPTH OF THE DAMAGE BY DRILLING A HOLE AND USING A SLIDE HAMMER TO PULL THE DENT OUT, OR INCREASE THE WIDTH OF THE DAMAGE AND THUS INCREASE THE FACTOR AND MAKE ALLOWABLE PER SRM 54-10-01 PAGE 104. I OPTED TO DO THE LATTER. IN OTHER WORDS, I POUNDED THE DENT AND MADE IT WIDER. AT THAT POINT THE DAMAGE WAS WITHIN SRM 54-10-01 LIMITS, AND THE DAMAGE WAS DEFERRED LONG TERM, AND THE COWL WAS TO BE REPLACED AT 'C' CHK. THE CAPT REFUSED THE ACFT, THE FLT CONSEQUENTLY CANCELED. ON MAR/XA/01 I FOUND OUT THAT THE ORIGINAL DAMAGE WAS ALLOWABLE (MAINT MANUAL 51-00-99 PAGE 201) UP TO 0.25 INCHES DEEP. BUT, WE HAD REWORKED THE AREA WHICH WAS IN VIOLATION OF MAINT MANUAL 51-70-01 PAGE 1, PARAGRAPH 2C. THE ACFT MAINT FERRIED (NON REVENUE) TO XYZ FOR REPLACEMENT OF THE NOSE COWL.

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## **SYNOPSIS**

A B737-300 WAS REPAIRED USING CONFLICTING AND CONFUSING SRM PROCS FOR NOW COWL DENT REWORK. PIC REFUSED ACFT FOR REVENUE FLT.

